

PRODUCT REQUIREMENTS DOCUMENT

LibraAI - University Library Research Assistant

Team 05 | Squad 69 | Sprint 2

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1. Executive Summary

University Library Research Assistant is a web-based research assistant designed to help students find concise, citation-backed answers from an assigned university library corpus. Instead of manually searching through research papers, theses, and course materials, students can ask natural-language questions and receive a short answer grounded only in retrieved library documents.

The system combines document ingestion, cleaning, section-aware chunking, semantic embeddings, vector retrieval, relevance thresholding, and constrained LLM generation. Every supported answer must include a traceable citation containing the source document and page or section. When the available corpus does not support an answer, the system explicitly responds that it does not know rather than guessing.

This PRD defines the product requirements, user needs, MVP boundaries, evaluation approach, risks, and acceptance criteria for Sprint 2.

2. Business Problem

2.1 Context

University libraries contain large collections of research papers, theses, and course materials. Although these documents contain the information students need, the information is distributed across many files and sections.

The current research experience is often:

- Manual — students must open and search many documents.
- Time-consuming — finding the exact supporting passage can take significant effort.
- Fragmented — relevant information may be distributed across multiple documents.
- Difficult to verify — students may find an answer without an immediately traceable source.
- Prone to unsupported answers — general-purpose AI can provide fluent information that is not present in the assigned corpus.

2.2 The Core Gap

There is a need for a focused intelligence layer over the university's assigned document corpus that connects natural-language questions with relevant document passages and produces concise answers with real citations.

The product must therefore connect:

Input	System Action	Output
Student question	Semantic retrieval across indexed corpus	Relevant document chunks
Relevant chunks	Relevance threshold check	Supported / unsupported decision
Supported chunks	Constrained LLM generation	Concise grounded answer
Retrieved sources	Metadata-based citation formatting	Document + page/section citation

2.3 Impact

- Students spend less time searching through unrelated documents.
- Answers are easier to verify because supporting sources are shown.
- Research across multiple documents becomes faster and more focused.
- The explicit refusal path reduces the risk of unsupported or hallucinated answers.
- The team gains a clear, explainable RAG pipeline suitable for demonstration and viva.

3. User Personas

3.1 Persona 1 — The Student Researcher

Attribute	Detail
Background	Primary user who is researching a topic, writing an assignment, or trying to understand a concept.
Primary Goal	Find a short, reliable answer from the university's available materials.
Frustration	Manually reading many papers to locate one answer.
Success	Can ask a question and quickly identify the answer and its exact supporting source.

3.2 Persona 2 — The Student Assignment Writer

Attribute	Detail
Background	Student who needs evidence-backed explanations while preparing coursework.
Primary Goal	Get concise explanations that can be traced back to a document and page/section.
Frustration	Difficulty verifying where an answer came from.
Success	Can expand citations and inspect the supporting chunk before using the information.

3.3 Persona 3 — Faculty / Mentor (Secondary)

Attribute	Detail
Background	Faculty member who may want to quickly check what the assigned library materials cover.
Primary Goal	Quickly determine whether a topic is covered and which materials discuss it.
Frustration	Searching across multiple course resources manually.
Success	Receives a grounded answer or an explicit indication that the corpus does not cover the question.

4. User Pain Points

ID	Persona	Pain Point	Severity
P1	Student Researcher	Cannot quickly locate the relevant passage among many documents.	Critical
P2	Student Researcher	Answers may require combining information from multiple documents.	High
P3	Assignment Writer	Cannot easily verify the exact page/section supporting an answer.	Critical
P4	All primary users	General AI answers may include information outside the assigned corpus.	Critical

P5	All primary users	Irrelevant or weak search results increase research time.	High
P6	All primary users	Scanned or inconsistently formatted documents may reduce retrieval quality.	High
P7	Students	Long documents can lose context when split using naive fixed-size chunks.	High
P8	Team	Changing metadata or chunking decisions after indexing can require re-indexing.	Medium

5. Project Goals

5.1 Primary Goals

Goal	Description
G1 — Fast Research	Allow students to ask natural-language questions instead of manually searching documents.
G2 — Grounded Answers	Generate answers using only content retrieved from the assigned corpus.
G3 — Traceable Citations	Attach document title and page/section information to supported claims.
G4 — Cross-Document Retrieval	Retrieve relevant chunks from multiple documents for one question.
G5 — Safe Refusal	Return an explicit “I don't know” response when corpus evidence is insufficient.
G6 — Explainable Pipeline	Keep ingestion, chunking, retrieval, prompting, and citation logic simple enough to explain in a viva.

5.2 Secondary Goals

Goal	Description
G7 — Source Inspection	Allow users to expand a citation and inspect the supporting chunk.
G8 — Basic Filtering	Optionally scope searches by course or document type.
G9 — Session Continuity	Optionally retain chat history during the current session only.

6. Corpus & Data Source Documentation

The assigned corpus is expected to contain research papers, theses, and course materials. The corpus must be inventoried before implementation so the team understands formats, document length, and extraction quality.

6.1 Corpus Inventory Requirements

- Count the total number of documents.
- Record formats such as PDF, DOCX, and HTML where applicable.

- Measure or estimate average document length.
- Identify scanned PDFs versus digital-native documents.
- Manually inspect 5–10 representative documents before the ingestion pipeline is finalized.
- Identify headers, footers, references/bibliographies, figures, tables, and other content that may affect retrieval.

6.2 Processing Pipeline

Stage	Requirement
Loading	Use PyMuPDF / unstructured to extract document content.
Cleaning	Remove or normalize extraction noise such as repeated headers/footers where appropriate.
Chunking	Use section-aware chunks of approximately 300–500 tokens with about 15% overlap.
Embedding	Use OpenAI text-embedding-3-small, or bge-small-en if a local/free alternative is required.
Indexing	Store embeddings and chunk metadata in ChromaDB.
Retrieval	Embed the student question and retrieve top-k semantically relevant chunks across the corpus.
Generation	Inject only retrieved content into the LLM prompt and require grounded output.

6.3 Data Quality Requirements

- Metadata must be defined before the first full embedding run.
- Page numbers and section information must be preserved whenever available.
- Malformed or unreadable documents must be logged and flagged rather than silently ignored.
- Bibliography/reference sections should be excluded or separately tagged when they create retrieval noise.
- The team must manually validate sample chunks before indexing the full corpus.

7. Success Metrics

7.1 Product Metrics

Metric	Target	Measurement Method
Answer Groundedness	≥ 80% on ground-truth evaluation set	Manual evaluation against known answers/sources
Retrieval Recall	≥ 80% target	Check whether the correct chunk appears in top-k
Citation Accuracy	100% of supported answers cite a real source	Compare displayed citation metadata with indexed source
Refusal Accuracy	Correct refusal on out-of-corpus test questions	Evaluation using deliberately unsupported questions
Query Latency	Target < 8–10 seconds	Measure end-to-end response time
Source Traceability	Every answer has document + page/section where available	UI and metadata inspection

7.2 Technical Metrics

Metric	Target
Ground-truth test set	10–15 questions prepared before development
Regression testing	Run test set after major chunking, retrieval, prompt, or model changes
Corpus indexing	Full assigned corpus indexed before final evaluation

Pipeline explainability	Each team member can explain the major RAG stages
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8. Functional Requirements

8.1 Query & Chat Module

ID	Requirement	Priority
FR-01	The system SHALL provide a chat-style interface for free-text student questions.	Must Have
FR-02	The system SHALL accept natural-language questions without requiring keyword-only search.	Must Have
FR-03	The system SHALL preserve chat history within the current session when session history is enabled.	Could Have

8.2 Retrieval Module

ID	Requirement	Priority
FR-04	The system SHALL embed the user's question using the configured embedding model.	Must Have
FR-05	The system SHALL retrieve the top-k relevant chunks across the entire indexed corpus.	Must Have
FR-06	The system SHALL retain source metadata with every retrieved chunk.	Must Have
FR-07	The system SHOULD support filtering by course or document type.	Could Have

8.3 Relevance & Refusal Module

ID	Requirement	Priority
FR-08	The system SHALL compare retrieval results against a defined relevance threshold.	Must Have
FR-09	If no retrieved chunk meets the threshold, the system SHALL return an explicit "I don't know" / "not covered in the available materials" response.	Must Have
FR-10	The system SHALL NOT generate an answer using unsupported external knowledge when corpus evidence is insufficient.	Must Have

8.4 Answer Generation Module

ID	Requirement	Priority
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FR-11	The system SHALL generate a concise answer using only the retrieved content.	Must Have
FR-12	The system SHALL instruct the LLM not to invent facts or citations.	Must Have
FR-13	The system SHALL include citation information for supported claims.	Must Have
FR-14	The system SHOULD indicate when multiple documents contribute to an answer.	Should Have

8.5 Citation & Source Module

ID	Requirement	Priority
FR-15	Every supported answer SHALL display at least one citation.	Must Have
FR-16	A citation SHALL contain the source document title and page number or section where available.	Must Have
FR-17	The user SHOULD be able to expand a citation and inspect the retrieved source chunk.	Should Have
FR-18	The system SHALL use stored metadata rather than generating citation details from the LLM's imagination.	Must Have

8.6 Backend & UI Module

ID	Requirement	Priority
FR-19	The backend SHALL expose the query workflow through FastAPI.	Must Have
FR-20	The frontend SHALL provide a simple chat interface using Streamlit by default, or Next.js if bandwidth allows.	Must Have
FR-21	The UI SHALL clearly distinguish an answer, citations, and refusal messages.	Must Have
FR-22	The UI SHOULD provide basic course/document-type filters if implemented within sprint capacity.	Could Have

9. Non-Functional Requirements

9.1 Performance

ID	Requirement
NFR-01	End-to-end query response should target under approximately 8–10 seconds for a live demo.
NFR-02	Retrieval should operate over the full indexed corpus rather than a manually selected document subset.

9.2 Reliability & Groundedness

ID	Requirement
NFR-03	The system must not fabricate citations.
NFR-04	Low-confidence retrieval must trigger the refusal path.
NFR-05	Ingestion failures must be logged without silently corrupting the index.

9.3 Maintainability & Explainability

ID	Requirement
NFR-06	Pipeline stages must be modular and understandable: ingestion, cleaning, chunking, embedding, retrieval, generation, citation.
NFR-07	Configuration such as model names, thresholds, and paths should be centralized.
NFR-08	Code should follow clear Python project conventions and include useful documentation.

9.4 Security

ID	Requirement
NFR-09	No API keys or secrets shall be committed to the repository.
NFR-10	Sensitive configuration shall be stored in environment variables / .env files excluded from version control.

9.5 Usability

ID	Requirement
NFR-11	A student should be able to submit a question without technical training.
NFR-12	Citations and refusal messages must be visually clear and easy to understand.
NFR-13	Source chunks should be expandable without obscuring the main answer.

10. User Stories

Epic 1 — Question Answering

US-101 — As a student researcher, I want to ask a natural-language question, so that I can get a concise answer without searching every document.

Acceptance Criteria:

- A question can be entered in the chat interface.
- The system searches the indexed corpus.
- The response is concise and grounded in retrieved material.

Epic 1 — Question Answering

US-102 — As a student, I want one question to search multiple documents, so that I can understand a topic covered across several sources.

Acceptance Criteria:

- Retrieval is performed across the full corpus.
- Relevant chunks may come from multiple documents.
- The final response identifies the supporting sources.

Epic 2 — Citations

US-201 — As an assignment writer, I want every answer to include a citation, so that I can verify the information.

Acceptance Criteria:

- At least one real source is shown for a supported answer.
- Citation includes document title and page/section when available.
- Citation metadata matches the indexed source.

Epic 2 — Citations

US-202 — As a student, I want to inspect the source chunk, so that I can understand the context behind the answer.

Acceptance Criteria:

- Citation can be expanded when the feature is implemented.
- The displayed chunk is the actual retrieved text.
- Source metadata is visible.

Epic 3 — Safe Refusal

US-301 — As a student, I want the system to say it does not know when the library does not cover my question, so that I do not mistake an unsupported answer for a reliable one.

Acceptance Criteria:

- An out-of-corpus test question triggers the refusal path.
- No unsupported factual answer is generated.
- The response clearly states that the available materials do not support an answer.

Epic 4 — Filtering

US-401 — As a student, I want to limit search to a course or document type, so that I can focus on relevant materials.

Acceptance Criteria:

- User can select a supported filter.
- Retrieval respects the selected scope.
- The filter is optional and does not prevent corpus-wide search.

Epic 5 — Session

US-501 — As a student, I want to see previous questions and answers during my session, so that I can continue my research conversation.

Acceptance Criteria:

- Current-session messages remain visible.
- History is not required to persist across separate sessions.

11. MVP Scope

11.1 Included in MVP

Feature	Rationale
Corpus inventory and sample inspection	Ensures the ingestion strategy matches real document formats.
Document ingestion and cleaning	Foundation for the retrieval system.
Section-aware chunking	Preserves context better than naive fixed-size splitting.
Embeddings + ChromaDB indexing	Enables semantic retrieval across the corpus.
Top-k cross-document retrieval	Core research-assistant capability.
Relevance threshold	Controls when the system is allowed to answer.
Grounded LLM answer generation	Produces concise natural-language responses.
Citation formatting	Makes every supported answer traceable.
Explicit refusal path	Prevents guessing when evidence is insufficient.
Streamlit chat UI + FastAPI backend	Primary live-demo delivery surface.
10–15 question evaluation set	Provides repeatable retrieval and answer-quality validation.

11.2 Explicitly Excluded from MVP

Feature	Reason for Deferral
Browser extension	Explicitly deferred until the core application is stable and mentor-approved.
Arbitrary external web/ebook ingestion	The sprint focuses on the assigned library corpus.
User accounts / login	Not required for sprint evaluation.
Personalization	Not required for the core research workflow.
Mobile application	Outside sprint scope.
Persistent cross-session history	Not required for the core deliverable.

12. Future Scope

Phase 2 — Enhanced Research Intelligence

- Browser extension for contextual research, subject to mentor sign-off.
- Advanced source navigation with direct document/page opening.
- Improved reranking and hybrid keyword + semantic retrieval.
- More sophisticated answer-level citation mapping.
- Persistent user workspaces and saved research sessions.
- Support for additional document formats and improved OCR workflows.

Phase 3 — Scale & Integration

- Role-based access for students and faculty.
- Institution-wide corpus management.
- Background indexing for newly added library documents.

- Analytics on frequently asked research topics.
- Production deployment with scalable vector storage and monitoring.

13. Risks and Assumptions

13.1 Risks

ID	Risk	Probability	Impact	Mitigation
R1	Scanned or poorly extracted PDFs reduce retrieval quality.	Medium	High	Inventory corpus early; time-box OCR/cleanup; flag major issues to mentor.
R2	Naive chunking loses section context.	Medium	High	Use section-aware 300–500 token chunks with overlap and manually validate samples.
R3	Reference sections pollute retrieval.	Medium	Medium	Detect and exclude or separately tag bibliographies/references.
R4	LLM produces unsupported statements.	Medium	High	Strict grounded prompt, relevance threshold, and refusal path.
R5	Similarity threshold is too strict or too loose.	Medium	High	Tune threshold using the 10–15 question ground-truth set.
R6	Citation metadata is incomplete.	Medium	High	Define metadata schema before embedding and preserve page/section during ingestion.
R7	API/model latency exceeds live-demo target.	Medium	Medium	Use efficient embedding/LLM models, caching where appropriate, and limit retrieved context.
R8	Team spends too long on document cleaning.	High	High	Time-box Week 2 work and prioritize the assigned corpus.
R9	Scope creep delays the core RAG pipeline.	High	High	Prioritize Must Have requirements and defer browser extension/integrations.

13.2 Assumptions

ID	Assumption
A1	The assigned document corpus is available to the team for ingestion and evaluation.
A2	The majority of documents can be processed into usable text, with OCR applied only where necessary.
A3	Python is the primary implementation language.

A4	ChromaDB is acceptable as the vector database for the sprint.
A5	OpenAI text-embedding-3-small is available; bge-small-en may be used if a local/free option is required.
A6	Claude or GPT-4o-mini is available for grounded answer generation.
A7	FastAPI and Streamlit are acceptable for the default backend/frontend architecture.
A8	The product is evaluated primarily as a live/demo application rather than a production library platform.

14. Acceptance Criteria

AC-1 — Corpus & Ingestion

- Assigned corpus is inventoried for document count, formats, length, and extraction quality.
- At least 5–10 representative documents are manually inspected before full indexing.
- Documents are cleaned and chunked using the agreed strategy.
- Each indexed chunk contains the required source metadata.

AC-2 — Retrieval

- A user question retrieves top-k relevant chunks from across the indexed corpus.
- The retrieval result contains document metadata and page/section information where available.
- The correct supporting chunk appears in top-k for at least 80% of the ground-truth questions (target).

AC-3 — Answer Generation

- Supported questions produce concise answers based only on retrieved content.
- The system does not intentionally use unsupported external facts.
- Answers meet the agreed target of at least 80% accuracy on the ground-truth evaluation set.

AC-4 — Citations

- Every supported answer displays at least one citation.
- Each citation identifies the source document and page/section when available.
- Citation details correspond to actual indexed metadata and are not fabricated by the LLM.

AC-5 — Refusal Path

- Out-of-corpus questions in the evaluation set trigger an explicit refusal.
- The refusal clearly communicates that the answer is not covered by the available materials.
- The system does not guess when no chunk meets the configured relevance threshold.

AC-6 — UI & Performance

- Chat interface accepts and displays questions and answers clearly.
- Citations are visually distinguishable from the main answer.
- Source chunks can be expanded if FR-17 is implemented.
- End-to-end query response targets approximately 8–10 seconds or less for the live demo.

AC-7 — Documentation & Viva Readiness

- Architecture and major pipeline decisions are documented.
- The team can explain ingestion, chunking, embeddings, retrieval, thresholding, generation, and citation flow.

- The evaluation set is version-controlled and rerun after major pipeline changes.

Appendix A — Glossary

Term	Definition
RAG	Retrieval-Augmented Generation — a workflow that retrieves source content before generating an answer.
Corpus	The assigned collection of library documents used by the assistant.
Chunk	A smaller piece of a document stored and retrieved independently.
Embedding	A numerical vector representation used to compare semantic similarity.
Vector Database	A database optimized for storing and searching vector representations.
Top-k Retrieval	Returning the k highest-ranked chunks for a query.
Relevance Threshold	Minimum similarity/relevance level required before the system may answer.
Grounded Answer	An answer supported by the retrieved corpus content.
Citation	Traceable source information identifying the document and page/section supporting an answer.
Refusal	An explicit response that the available corpus does not contain enough information to answer.
OCR	Optical Character Recognition used to extract text from scanned documents.

Appendix B — Metadata Schema Contract

The metadata schema should be finalized before the first embedding run.

Field	Description
doc_title	Title of the source document.
doc_type	Document category such as research paper, thesis, or course material.
author	Author name(s), when available.
course_code	Associated course code, when applicable.
section	Document section containing the chunk, when available.
page_number	Source page number containing the chunk, when available.

Recommended additional fields:

Field	Purpose
chunk_id	Unique identifier for the indexed chunk.
source_path	Internal path or identifier for the original document.
text	The cleaned chunk content stored with the vector record or retrievable document store.